

Gravitational Engineering and Interstellar Transit in the Informational Actualization Model: A First-Principles Exploration of What the Framework Permits

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Status of this document. This is an exploratory note, not a validation paper. It is divided into two explicitly separated zones. **Zone I** contains only results that follow rigorously from established physics (general relativity, the equivalence principle, expansion kinematics) combined with the already-stated identifications of the Informational Actualization Model (IAM). Every equation in Zone I is derivable and is either standard physics or follows from IAM’s published structure. **Zone II** is conjectural. It begins at an explicitly marked transition and concerns a regime that requires a material whose existence is not established. No equation in Zone II is claimed as derived; each is presented as the *form a future derivation would have to take*, conditional on premises that are stated as premises. The boundary between what is established and what is conjectured is never blurred. This separation is the entire point of the document.

Premise: what IAM already says about gravity

IAM identifies gravity not as a fundamental force but as the local manifestation of an information-theoretic process. At each irreversible quantum-to-classical transition (decoherence) on a timelike worldline, classical information is produced and paid for at the Landauer cost

$$\Delta E_{\text{act}} = k_B T_H \ln 2 \quad \text{per bit}, \quad T_H = \frac{\hbar H}{2\pi k_B}, \quad (1)$$

where T_H is the (Gibbons–Hawking) horizon temperature. This cost is partitioned by the virial theorem into two equal halves: a geometric/potential half that manifests locally as the gravitational potential, and a kinetic/informational half that manifests globally as the dark-energy term in the expansion. The partition is fixed at 1/2 by the equilibrium condition (the coupling constant $\beta_m = \Omega_m/2$), and the cumulative actualization is encoded in the activation function

$$\mathcal{E}(a) = \exp\left(1 - \frac{1}{a}\right), \quad \mathcal{E}(1) = 1, \quad \mathcal{E}(a \rightarrow \infty) = e, \quad (2)$$

entering the matter-sector expansion as

$$H_{\text{eff,m}}^2(a) = H_{\Lambda\text{CDM}}^2(a) + \beta_m \mathcal{E}(a) H_0^2. \quad (3)$$

These are not new to this document; they are the established core of IAM and are tested against data elsewhere. The present note asks a single question: *taking these as given, what does the framework permit?*

ZONE I — Established physics and IAM consequences

I.1 Gravity as a steerable effective potential

In IAM the locally felt gravitational acceleration is the gradient of an effective potential built from the potential half of the partition:

$$\mathbf{g}_{\text{IAM}}(\mathbf{x}) = -\nabla\Phi_{\text{IAM}}(\mathbf{x}). \quad (4)$$

This is definitional and recovers Newtonian gravity in the weak field. The content of Zone I is the following: *whatever physical process sets Φ_{IAM} , a body responds only to its local gradient.* Equation (4) is the only ingredient required for the kinematic results below. We do not yet ask how Φ_{IAM} might be engineered; that is Zone II. We ask only what follows *if* the gradient can be shaped.

I.2 Free-fall motion carries no felt acceleration (equivalence principle)

A body moving solely under $\mathbf{g}_{\text{IAM}}$ follows a geodesic. Its equation of motion is

$$\ddot{\mathbf{x}} = -\nabla\Phi_{\text{IAM}}(\mathbf{x}). \quad (5)$$

Every part of the body, and any occupant, obeys the same equation. By the equivalence principle (established physics since 1907), an observer in free fall experiences *no* proper acceleration: the locally felt force is

$$\mathbf{f}_{\text{felt}} = m(\ddot{\mathbf{x}} + \nabla\Phi_{\text{IAM}}) = 0, \quad (6)$$

identically, regardless of the magnitude or direction of $\nabla\Phi_{\text{IAM}}$, provided the gradient is uniform across the body's scale L . The residual tidal force is the differential of the gradient,

$$\mathbf{f}_{\text{tidal}} \sim m L \partial^2\Phi_{\text{IAM}}, \quad (7)$$

and vanishes in the limit of a gradient uniform over L .

Consequence (rigorous). A craft that moves by following a shaped Φ_{IAM} gradient undergoes *no felt acceleration during arbitrary changes of speed or direction.* A 90° “turn” is accomplished by relocating the gradient direction; the craft and occupants remain in free fall throughout and feel nothing. This dissolves the inertial objection to abrupt high-acceleration maneuvers. It is a direct consequence of the equivalence principle and requires no new physics.

I.3 The hover and the abrupt departure as gradient operations

A stationary hover corresponds to shaping Φ_{IAM} so that its gradient cancels the ambient gravitational gradient,

$$\nabla\Phi_{\text{IAM}} = -\mathbf{g}_{\text{ambient}}, \quad (8)$$

leaving the craft in a local null. An abrupt departure corresponds to a steep gradient placed along the departure direction; the craft free-falls into it with coordinate acceleration $|\nabla\Phi_{\text{IAM}}|$ but, by I.2, zero felt acceleration. Both are the same operation (shaping the gradient) at different settings. No equation here exceeds Newtonian gravity plus the equivalence principle.

I.4 Orientation along the gradient: the tilt signature

A rigid body suspended in a gradient orients so that its constrained axis aligns with the local “down” defined by $-\nabla\Phi_{\text{IAM}}$, exactly as a plumb bob hangs toward the net field. If forward motion is produced by placing the gradient focus forward and below the craft, the craft’s equilibrium orientation tilts toward that focus by an angle

$$\theta_{\text{tilt}} = \arctan\left(\frac{|\nabla\Phi_{\text{IAM}}|_{\text{horizontal}}}{|\nabla\Phi_{\text{IAM}}|_{\text{vertical}}}\right). \quad (9)$$

Because the horizontal gradient component grows with the commanded acceleration, θ_{tilt} is a monotonically increasing function of acceleration magnitude.

Falsifiable signature (Zone I). A gradient-following craft must *tilt toward its direction of acceleration*, with tilt angle increasing with acceleration, and must *not* bank (roll about the velocity axis) as an aerodynamic vehicle does. The discriminator is kinematic and is in principle testable against motion data: lean correlated with acceleration magnitude, in the direction of acceleration, with no aerodynamic banking roll, distinguishes a gradient-follower from a lift-based aircraft. This prediction stands independent of any claim about how the gradient is produced.

I.5 Expansion kinematics: the relevant “speed limit” is not c

The cosmological expansion already separates comoving objects at recession speeds

$$v_{\text{rec}} = H_0 D, \quad (10)$$

which exceeds c for $D > D_H$, the Hubble radius,

$$D_H = \frac{c}{H_0} \approx \frac{c}{70 \text{ km s}^{-1} \text{ Mpc}^{-1}} \approx 1.4 \times 10^{10} \text{ light-years}. \quad (11)$$

No object moves through space faster than c ; space itself carries them. This is standard cosmology. In IAM the expansion is the global expression of the kinetic/informational half of the partition, Eq. (3). Thus the framework already contains a mechanism by which proper separations grow faster than c without any local violation of relativity. The “speed limit” relevant to a space-carrying drive is therefore not c ; it is set by how steep an actualization imbalance can be sustained (developed in Zone II), bounded above by the asymptote $\mathcal{E} \rightarrow e$ of Eq. (2).

I.6 Proper time aboard a locally inertial bubble

Consider a craft enclosed in a region of locally flat spacetime that is itself displaced relative to distant stars by reshaping the surrounding geometry (the mechanism is Zone II; the kinematics here are not). Inside a locally flat region the craft is inertial, so the proper time elapsed aboard equals the coordinate time of the bubble’s worldline,

$$d\tau_{\text{ship}} = dt_{\text{bubble}}, \quad (12)$$

and because the craft never acquires large *local* velocity through space, it accumulates none of the special-relativistic dilation factor $\sqrt{1 - v^2/c^2}$ that afflicts a near- c through-space voyage. Consequently the elapsed time on the craft and the elapsed time at the (comoving) origin differ only by the second-order gravitational-potential term associated with the engineered region,

$$\frac{\Delta\tau_{\text{Earth}} - \Delta\tau_{\text{ship}}}{\Delta\tau} \sim \frac{\Delta\Phi_{\text{IAM}}}{c^2} \ll 1 \quad (13)$$

for any potential small compared with c^2 .

Consequence (rigorous, conditional on a flat bubble existing). A round trip that takes proper time $\Delta\tau_{\text{ship}}$ aboard elapses approximately the same $\Delta\tau_{\text{Earth}}$ at the origin — in sharp contrast to relativistic through-space travel, where the twin asymmetry makes $\Delta\tau_{\text{Earth}} \gg \Delta\tau_{\text{ship}}$. This is the Alcubierre-class result: a space-carrying displacement does not incur the twin-paradox penalty because the travellers remain locally inertial. It follows from the metric structure of a locally flat displaced region and requires no new physics beyond the existence of such a region.

I.7 The transit figure for the Pleiades

The Pleiades lie at $D \approx 136 \text{ pc} \approx 444 \text{ light-years}$. For a space-carrying displacement at effective rate $v_{\text{eff}} = \xi c$ (with ξ the ratio of induced proper-separation rate to c ; $\xi > 1$ is permitted because no local velocity exceeds c), the one-way coordinate time is

$$t_{\text{one-way}} = \frac{D}{v_{\text{eff}}} = \frac{444 \text{ ly}}{\xi c} = \frac{444}{\xi} \text{ years.} \quad (14)$$

By I.6 the proper time aboard is approximately equal. A two-week round trip ($\Delta\tau_{\text{ship}} \approx 14 \text{ days}$, so $t_{\text{one-way}} \approx 7 \text{ days} \approx 0.0192 \text{ yr}$) requires

$$\xi = \frac{444 \text{ yr}}{0.0192 \text{ yr}} \approx 2.3 \times 10^4. \quad (15)$$

That is, the surrounding geometry must be reshaped so that proper separation changes at $\sim 2 \times 10^4 c$ -equivalent. **This number is not forbidden by relativity** (it is a rate of proper-separation change, not a local velocity), and it is far below the regime where the actualization asymptote $\mathcal{E} \rightarrow e$ would impose a ceiling: 444 light-years is a negligible fraction of the Hubble radius $D_H \sim 1.4 \times 10^{10} \text{ ly}$ of Eq. (11). *The distance to the Pleiades is not the obstacle.* What remains is the question of whether such a geometry can be engineered — which is Zone II.

TRANSITION TO ZONE II — conjecture begins here

Everything above follows from established physics plus IAM's existing structure. **Nothing below is derived.** The remainder of this document concerns the engineering coupling — how an external apparatus might *write* to the actualization field Φ_{IAM} rather than merely read it — and the material that would be required. This coupling is *not present in IAM as currently formulated*; it is the open derivation. Worse, it is conjectured to require a material (a long-lived island-of-stability superheavy held in macroscopic quantum coherence) whose existence is not established. **We therefore cannot derive this section.** We state only the *form* that the missing physics would have to take, clearly labelled as conjecture, so that the structure of the open problem is recorded. Each numbered item below is a premise or a conditional, never a result.

ZONE II — the conjectural engineering coupling

II.1 The missing object: an actualization coupling

Conjecture C1. There exists a coupling by which a controlled physical input $J(\mathbf{x}, t)$ sources a local perturbation of the actualization field:

$$\Phi_{\text{IAM}}(\mathbf{x}, t) = \Phi_0(\mathbf{x}) + \int G(\mathbf{x} - \mathbf{x}', t - t') J(\mathbf{x}', t') d^3x' dt' \quad (\text{conjectured form, not derived}), \quad (16)$$

with G a response kernel. *The kernel G is unknown.* IAM specifies that Φ_{IAM} is sourced by the rate of irreversible actualization; C1 conjectures that this rate can be *driven* rather than only observed. Deriving G from the partition structure is the central open problem and is not attempted here.

II.2 The material premise

Premise P1 (knife-edge). Driving the actualization rate requires a substrate whose mass-energy is metastable with respect to the potential/actual boundary — poised, so a small input tips a large mass-energy across in a controlled way. Ordinary matter is fully actualized and does not respond. The island of stability (superheavy nuclei near the predicted shell closures, $Z \sim 114\text{--}126$) is the conjectured location of such metastable matter.

Premise P2 (coherence). To hold an induced imbalance against the restoring tendency of the partition (the equilibrium that fixes the 1/2 split), and to suppress the substrate’s ordinary gravity-sourcing, the material must maintain macroscopic quantum coherence — i.e. be superconducting in the operational state.

Tension. P1 and P2 are in conflict in ordinary materials science: nuclear metastability and macroscopic electronic coherence are not known to coexist. The conjecture is that the shell-closure structure granting the island its longevity may also support the ordered coherent state — i.e. that the same magic numbers enable both. *This is unproven and may be false.* **Because this material is not known to exist, no quantitative derivation past this point is possible,** consistent with the principle that one cannot derive physics that requires an unestablished substrate.

II.3 Predicted signature of the material, if it exists

Conditional prediction. If P2 holds, a portion of the material’s mass-energy is held in a coherent, “not-yet-actualized” state and does not source gravity in proportion to its rest mass. Such a sample would exhibit an *anomalous gravitational/inertial signature* — it would weigh slightly less than its measured mass-energy predicts. This is the one element of Zone II that is empirically checkable should such a material ever be produced: *weigh it; if it is gravitationally anomalous, P2 is supported; if not, the conjecture fails.*

II.4 The three-source steering geometry (conditional architecture)

Conditional on C1 and a responsive material, the focal placement geometry follows from elementary field superposition. Three non-collinear sources at positions $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3$, each driven with amplitude A_i and phase ϕ_i , produce a combined perturbation whose focal node $\mathbf{x}_f$ is steerable throughout the 3D volume by adjusting the relative $\{A_i, \phi_i\}$:

$$\Phi_{\text{drive}}(\mathbf{x}) = \sum_{i=1}^3 A_i G(\mathbf{x} - \mathbf{r}_i) e^{i\phi_i} \quad (\text{conditional on the unknown } G). \quad (17)$$

Three is the minimum number of non-collinear sources whose relative amplitudes span placement of a single focal point throughout a 3D region; two give only axial placement, and the monopole of one gives none. The coherence premise P2 is what permits a *stable phase relation* among the three sources, hence a sharp (rather than noise-averaged) focal node. This is why the three elements — material, sources, coherence — are jointly necessary and individually insufficient: the material supplies tippable mass-energy, the three sources supply steerable aim, and coherence couples the aim to the reservoir as a single phase-locked unit.

II.5 What would close the gap

The single derivation that would move the interstellar result from Zone II to Zone I is the response kernel G of Eq. (16), derived from the IAM partition structure, together with a demonstration that a material satisfying P1 and P2 can exist. Absent either, the kinematic results of Zone I stand on their own — they state, rigorously, that *if* the geometry can be shaped, the ride is weightless, the twin penalty is avoided, and 444 light-years is not the obstacle — while the engineering remains conjectural. The honest status of the whole is therefore: *the destination is permitted by the kinematics; the vehicle is not yet derivable.*

Summary

Zone I (established). Given IAM’s identification of gravity as the gradient of an actualization potential: a gradient-following craft feels no acceleration during arbitrary maneuvers (equivalence principle); hover, abrupt departure, and the tilt-toward-acceleration signature follow as gradient operations; the cosmological expansion already carries proper separations faster than c without local violation; a locally flat displaced region avoids the twin-paradox penalty; and the 444 light-years to the Pleiades is a negligible fraction of the Hubble radius, requiring an effective rate $\xi \sim 2 \times 10^4$ that relativity does not forbid. These are consequences of existing physics plus IAM’s existing structure.

Zone II (conjecture). The engineering coupling that would let an apparatus drive (not merely observe) the actualization field is not derived and is not present in IAM as formulated; it is conjectured to require a metastable, macroscopically coherent superheavy material whose existence is unestablished, with the one checkable prediction that such a material would weigh less than its mass-energy implies.

The boundary between these two zones is the honest content of this note. The kinematics say the stars are not as far as they look. The engineering says we cannot yet show how to go. Both statements are true at once, and keeping them separate is the point.

Exploratory note. Zone I derivable from established physics and IAM structure; Zone II explicitly conjectural. Timestamped in repository.